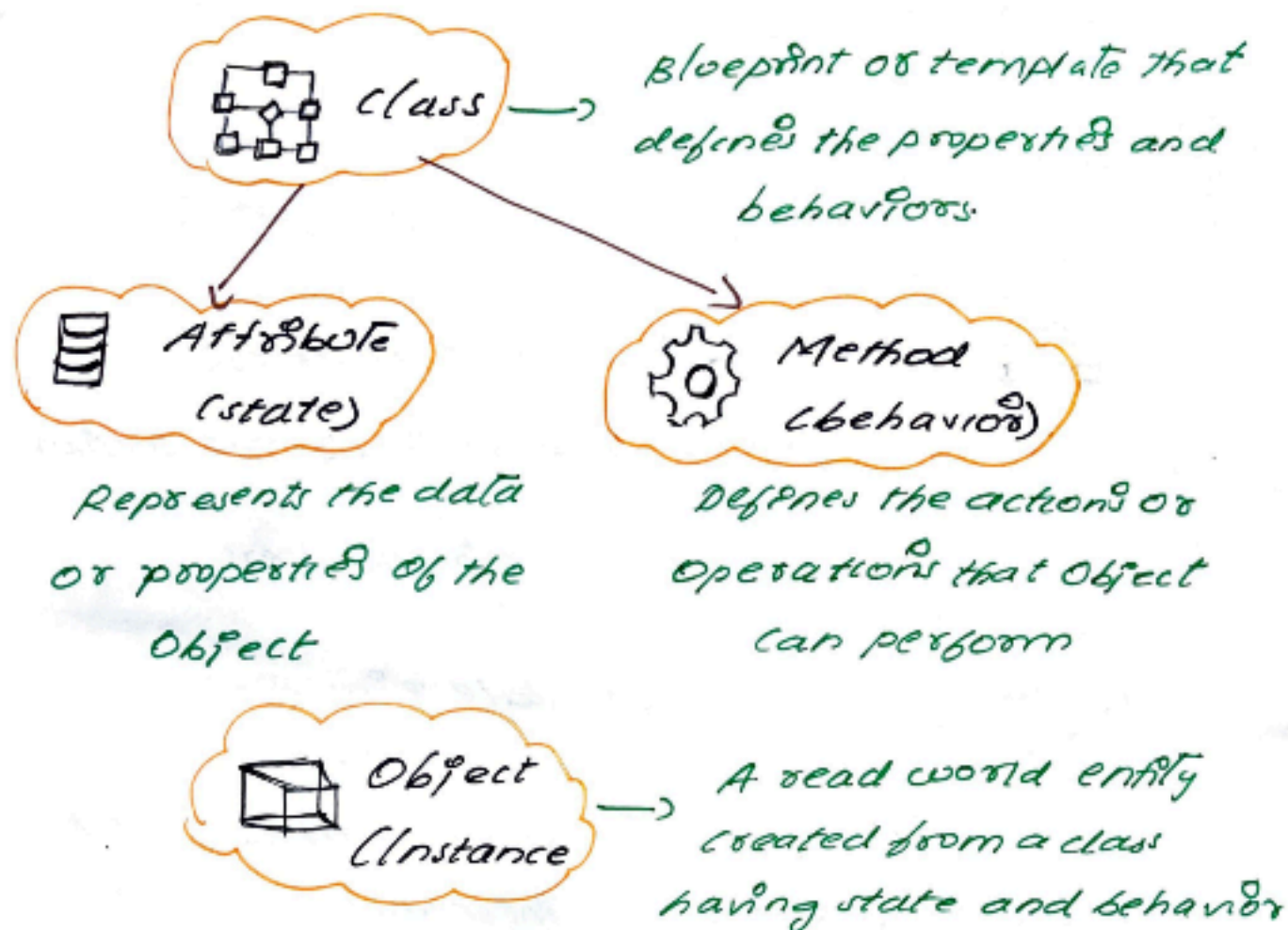


### 3. class, Object, method & attribute



### 4. Software Quality



## So UML Diagrams classification

UML (Unified modeling language.)

Structural diagrams



- class diagram
- object diagram
- component diagram
- package diagram

Behavioral diagrams



- use case diagram
- activity diagram
- state machine diagram

Interaction diagram



- sequence diagram
- communication diagram
- interaction overview diagram

Benefits: Visualization, Better communication, Better analysis.

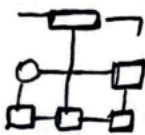
## 6. Failure during coding phase.

Requirement Design Coding Testing Deployment



Reason

- Implementation errors in program logic, syntax or algorithm



Impact

- errors propagate to next phase
- Testing becomes ineffective

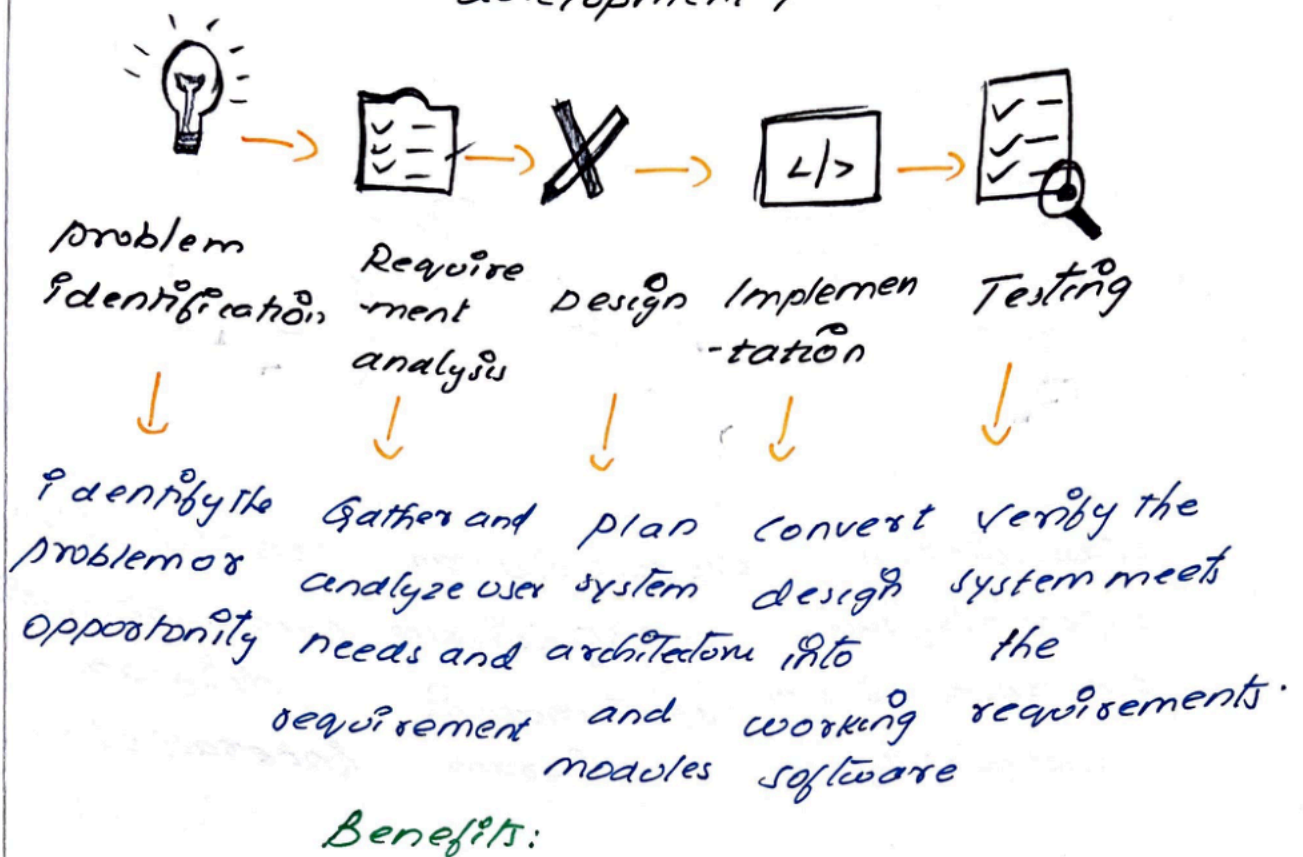


Solution

- Review source code
- follow coding
- conduct code inspection before testing

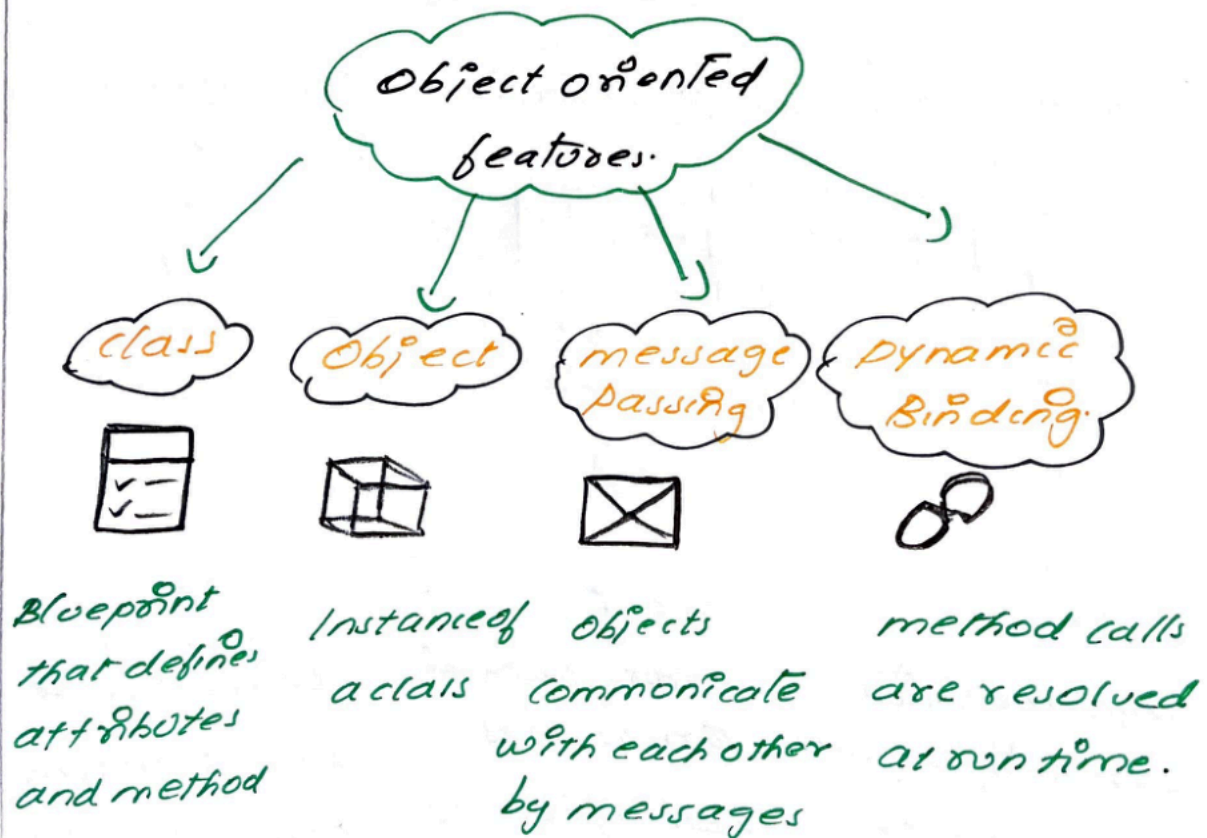


## 7. software development process



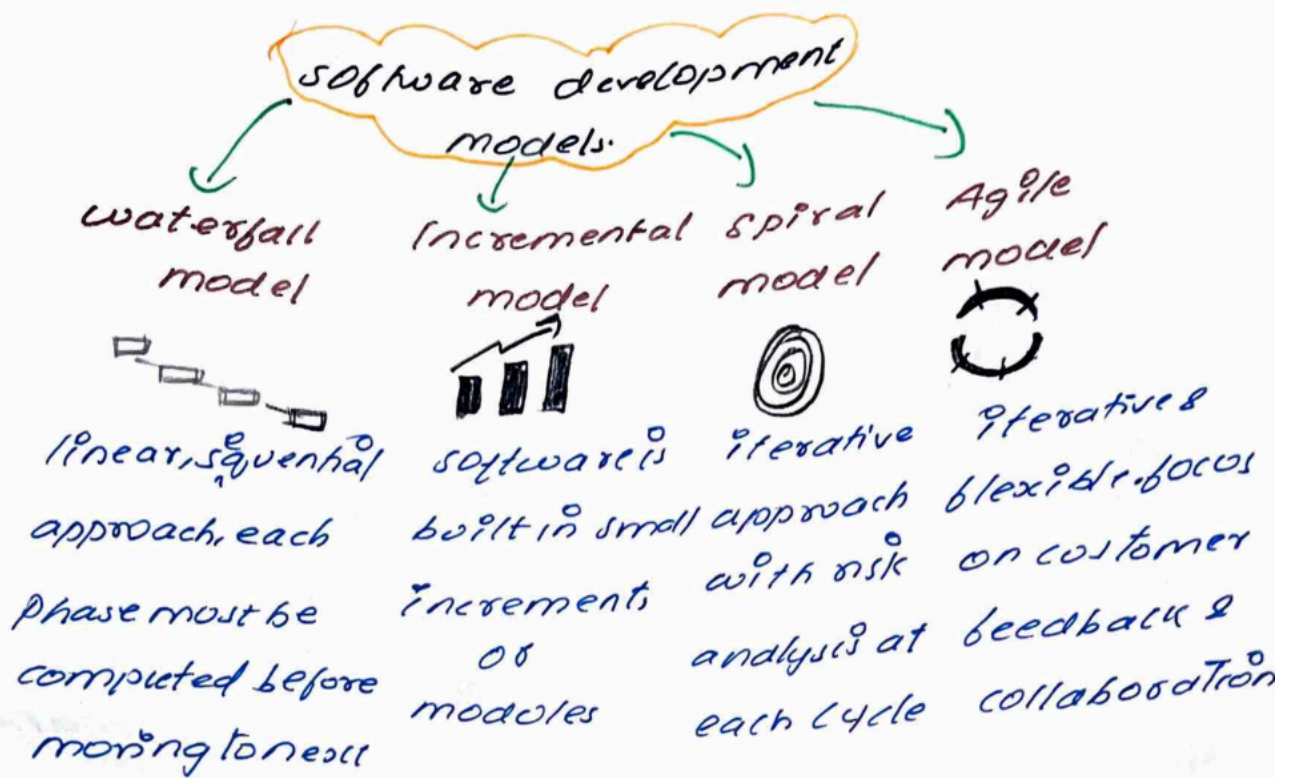
- ✓ Improves quality
- ✓ Reduces errors
- ✓ ensure success.

## 8. Object Oriented features.



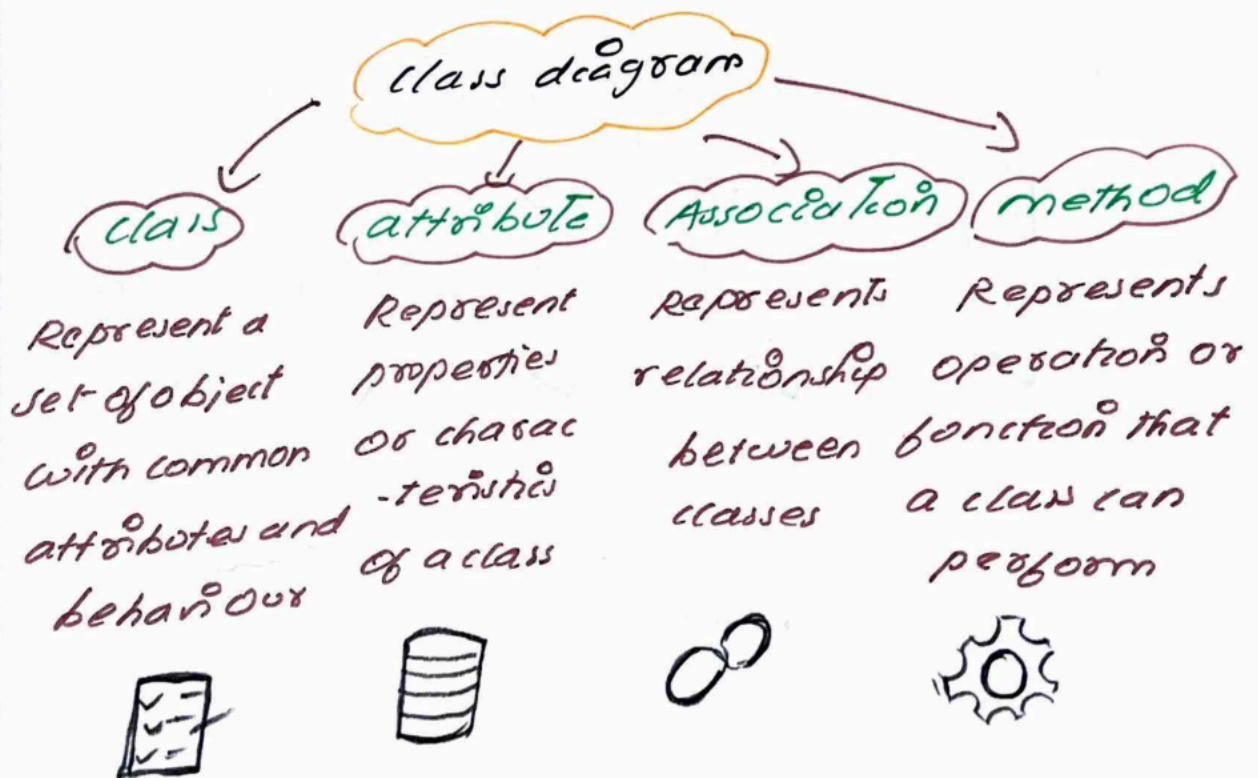
**Result:** Flexibility, maintainability, Runtime polymorphism documentation.

## 9. Software Development models



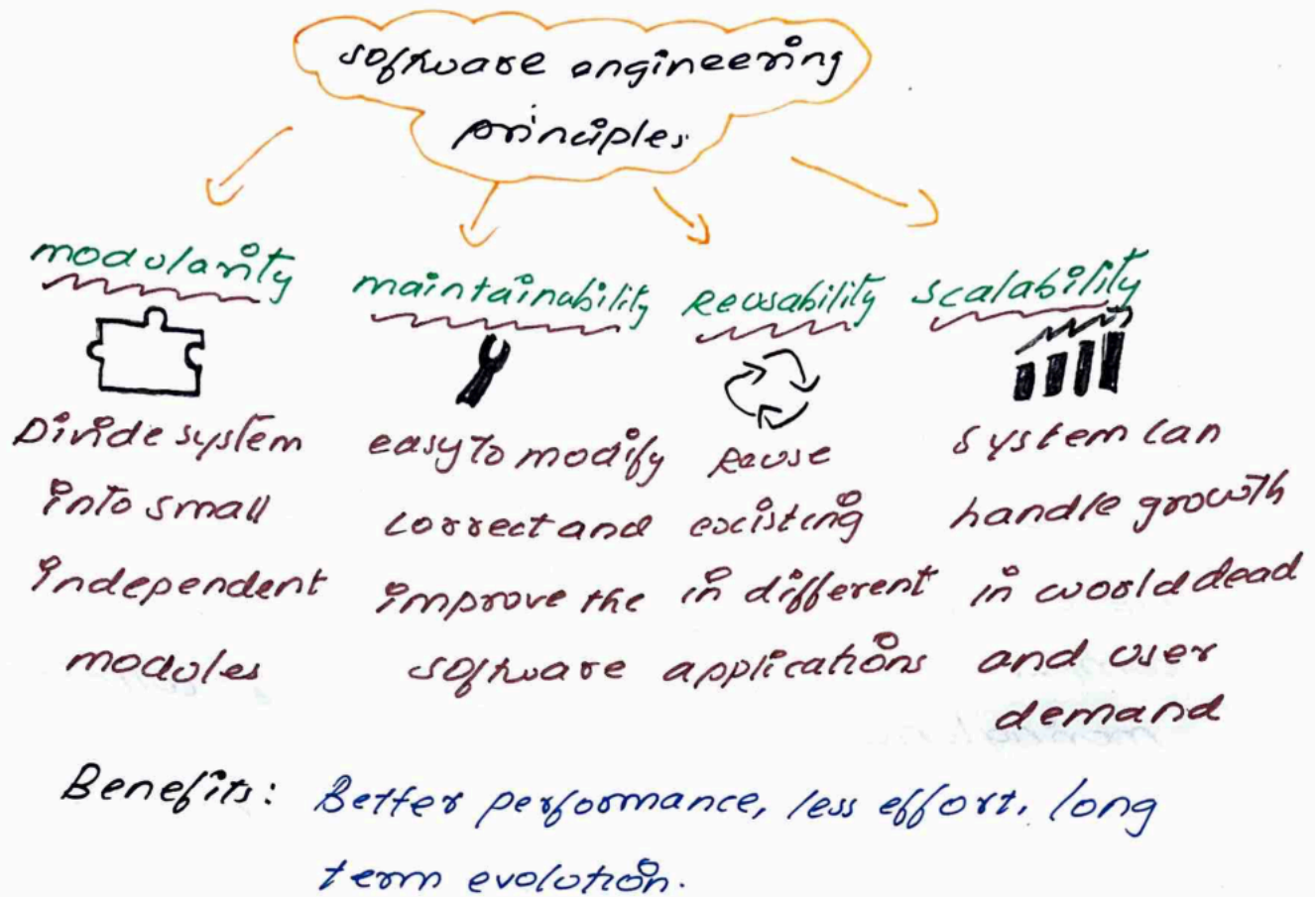
**Benefits:** Adaptability, risk reduction, Customer Satisfaction

## 10. Class diagram elements



**Benefits:** Structure Representation, Better documentation, supports design.

## 11. software engineering principles



## 12. failure during Integration phase.

